

Linyun He

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Education

- **Ph.D. in Operations Research, Minor in Mathematics** 2025
Advisor: Dr. Eunhye Song
Georgia Institute of Technology Atlanta, GA
- **M.A. in Statistics** 2019
Columbia University New York, NY
- **B.S. in Mathematics and Applied Mathematics** 2017
Fudan University Shanghai, China

Academic position

- **Postdoctoral Research Fellow** 2025-Now
Supervisors: Dr. Carolina Osorio (HEC Montreal) and Dr. Fausto Errico (ETS)
HEC Montreal Montreal, Canada

Research Domain

- Methodologies: simulation-based combinatorial optimization, sequential decision making, uncertainty quantification, nonparametric and high-dimensional statistics, simulation inference and optimization under model risk.
- Applications: regenerative simulation, smart manufacturing, digital twins, and transportation systems.

Publications

- [1] **Linyun He**, Luke Rhodes-Leader, and Eunhye Song (2024). Digital Twin Validation with Multi-Epoch, Multi-Variate Output Data. In *Proceedings of the 2024 Winter Simulation Conference (WSC)*, 347–358.
- [2] **Linyun He** and Eunhye Song (2024). Introductory Tutorial: Simulation Optimization under Input Uncertainty. In *Proceedings of the 2024 Winter Simulation Conference (WSC)*, 1338–1352.
- [3] **Linyun He**, Uday V Shanbhag, and Eunhye Song (2024). Stochastic Approximation for Multi-period Simulation Optimization with Streaming Input Data. *ACM Transactions on Modeling and Computer Simulation*, 34(2):1–27.
- [4] **Linyun He**, Eunhye Song, and Ben Feng (2023). Efficient Input Uncertainty Quantification for Regenerative Simulation. In *Proceedings of the 2023 Winter Simulation Conference (WSC)*, 385–396.
- [5] Zhunxuan Wang, **Linyun He**, Chunchuan Lyu, and Shay Cohen (2022). Nonparametric Learning of Two-Layer ReLU Residual Units. *Transactions on Machine Learning Research*, 1–41.
- [6] **Linyun He** and Eunhye Song (2021). Nonparametric Kullback-Liebler Divergence Estimation Using M-spacing. In *Proceedings of the 2021 Winter Simulation Conference (WSC)*, 1–12.
- [7] Zihao Wang, **Linyun He**, Zhenyun Qin, Roger Grimshaw, and Gui Mu (2019). High-order Rogue Waves and Their Dynamics of the Fokas–Lenells Equation Revisited: A Variable Separation Technique. *Nonlinear Dynamics*, 98:2067–2077.

Preprints and Working Papers

- [1] **Linyun He**, Ben Feng, and Eunhye Song (2025). Efficient Input Uncertainty Quantification for Ratio Estimator. Under Minor Revision at *INFORMS Journal on Computing*.
- [2] **Linyun He**, Luke Rhodes-Leader, and Eunhye Song (2025). Validating a Stochastic Digital Twin with Multi-Epoch, Multi-Variate Data. Submitted.

Awards and Honors

- [1] Winner of the 18th GERAD Postdoctoral Fellowship Competition 2025
- [2] Finalist of the 2023 Winter Simulation Conference Best Theoretical Contributed Paper Award (5/209) 2023
- [3] Overseas Exchange Scholarship by Fudan University Education Development Foundation 2017

- [4] Third Prize of the Scholarship for Outstanding Students at Fudan University 2014, 2015, 2016
- [5] Major Scholarship of School of Mathematical Sciences at Fudan University 2014, 2015, 2016

Conference Presentations

- [1] 2025 INFORMS Annual Meeting Atlanta, GA, 2025.10
Title: Simulation Digital Twin Validation with Multi-Epoch Key Performance Indicators
- [2] 2024 Winter Simulation Conference Orlando, FL, 2024.12
Title a: Digital Twin Validation with Multi-epoch, Multi-variate Output Data
Title b: Introductory Tutorial: Simulation Optimization under Input Uncertainty
- [3] 2024 INFORMS Annual Meeting Seattle, WA, 2024.10
Title: Digital Twin Validation with Multi-Epoch, Multi-Dimensional Data
- [4] 2023 Winter Simulation Conference San Antonio, TX, 2023.12
Title: Efficient Input Uncertainty Quantification for Regenerative Simulation
- [5] 2023 INFORMS Annual Meeting Phoenix, AZ, 2023.10
Title: Efficient Input Uncertainty Quantification for Regenerative Simulation
- [6] 2023 IISE Annual Conference New Orleans, LA, 2023.05
Title: Efficient Input Uncertainty Quantification for Steady-state Simulation
- [7] 2022 IISE Annual Conference Seattle, WA, 2022.05
Title: Statistical Analysis of Data-driven Simulation: Model Calibration and Optimization
- [8] 2021 Winter Simulation Conference Phoenix, AZ, 2021.12
Title: Nonparametric Kullback-Liebler Divergence Estimation Using M-Spacing
- [9] 2020 INFORMS Annual Meeting online, 2020.11
Title: Stochastic Approximation for Simulation Optimization with Streaming Input Data

Teaching Experience

- *Instructor* for ISYE 3044 Simulation Analysis and Design Georgia Tech, summer 2023
- *Teaching Assistant* for IE425 Stochastic Models in Operations Research Penn State Univeristy, spring 2021
- *Teaching Assistant* for IE453 Simulation Modeling For Decision Support Penn State Univeristy, fall 2020

Industrial Experience

- *Research Scientist Intern* at **Amazon.com Inc** Bellevue, Washington 2022.05-2022.08
- *Quantitative Analyst Intern* at **COFCO Futures Co., LTD** Shanghai, China 2017.01-2017.03
- *Quantitative Analyst Intern* at **Everbright Securities Co., LTD** Shanghai, China 2016.06-2016.08
- *Audit Intern* for **KPMG** Shanghai, China 2016.01-2016.02

Service

- Session Chair for 2024 INFORMS Annual Meeting
- Session Chair for 2023–2024 Winter Simulation Conferences

Software

- `alocv`: R package for computationally efficient approximation of the leave-one-out cross validation risk